ESP PREDICTIVE MAINTENANCE

Electric Submersible Pump Monitoring & Failure Prediction

DATABRICKS SOLUTION ACCELERATOR

Powered by Databricks Apps • Unity Catalog • Serverless

END-TO-END ML PIPELINE FOR ESP MONITORING & FAILURE PREDICTION

ESP Predictive Maintenance delivers a complete ML-powered platform — from telemetry ingestion through XGBoost model training, streaming inference, and operational dashboards — to predict and prevent costly ESP failures across upstream artificial lift systems.

◆ BUSINESS VALUE

- **Prevent unplanned failures** saving $100K–$500K per workover event in deferred production costs
- **Extend ESP run life** with physics-based diagnostics detecting motor overload, gas interference, and bearing wear
- **Optimize maintenance scheduling** with AI-driven work order prioritization and SAP ERP integration
- **Improve fleet visibility** with real-time health scores and failure risk across 12 wells in 5 basins
- **Accelerate root cause analysis** with Foundation Model API-powered troubleshooting assistant

◆ CHALLENGES ADDRESSED

- **Reactive maintenance** — ESP failures detected only after production impact and costly workovers
- **Fragmented diagnostics** — sensor data, work orders, and spare parts in disconnected systems
- **Limited predictive capability** — no ML models scoring telemetry in real time for early warning
- **Manual troubleshooting** — hours spent diagnosing faults that could be automated with AI
- **Inventory blind spots** — spare parts stockouts causing extended well downtime

◆ WHO IT'S FOR

🔧 **Production Engineer**
Real-time ESP health monitoring, failure risk scores, and diagnostic fault codes

📊 **Maintenance Manager**
Work order tracking, spare parts inventory, and planned-vs-reactive KPIs

🤖 **Data Scientist**
End-to-end ML pipeline with feature engineering, XGBoost, and MLflow tracking

🛢️ **Field Operations Lead**
Fleet-wide ESP status, live alerts, and AI-powered troubleshooting

12	5	6	8
Wells Monitored	Basins	Dashboard Tabs	Sensor Channels

◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the ESP-PM repo to your local machine or workspace
2. **Run DDL Scripts**
   Create Unity Catalog tables and Lakebase schema
3. **Deploy the App**
   Run databricks apps deploy to launch the dashboard
4. **Train & Score**
   Run the ML pipeline notebooks for XGBoost training

◆ GETTING THE MOST OUT OF ESP PREDICTIVE MAINTENANCE

📡 **Monitor Fleet Command**
Track all 12 ESPs with live health gauges, failure risk scores, and production KPIs

🔍 **Diagnose Wells**
Deep-dive into per-well sensor trends, diagnostic fault codes, and root cause analysis

🤖 **Ask ESP Advisor**
Chat with the Foundation Model API assistant for well-specific troubleshooting

⚙️ **Track SAP Maintenance**
Manage work orders, spare parts inventory, and maintenance KPIs

◆ TARGET INDUSTRY

Upstream Oil & Gas — Artificial lift operations across unconventional plays including Permian Basin, Eagle Ford, DJ Basin, Bakken, and Marcellus formations.

⚠ **DEMO ONLY** — This application is a demonstration and solution accelerator. Please refer to the included LICENSE file for full terms of use.

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